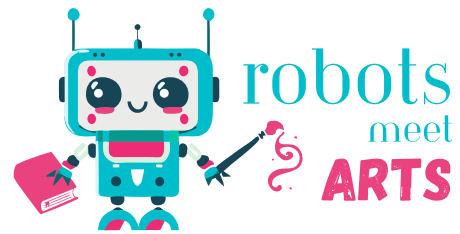




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000151881

Drop the bass!



Subject:
building a bass guitar
Content:

- Kids discover the basics of a bass guitar
- Build a bass guitar with makey makey

Computational practices:
choose what fits

- Decomposing
- Abstraction
- Problem solving

Unplugged/**plugged**



Preparation time:
15' min



Age:
10- 12 years



Duration:
100' min



Location:
classroom



Group Distribution:

- Solo participants or pairs to groups of 2-4

This depends on the number of Makey Makeys available.



Difficulty



A fantastic way to introduce this lesson is by using the Makey Makey piano.

Although it's a classic, it remains a favorite! You can find lesson plans and videos on the Makey Makey website. If the students construct the piano, they will already understand:

- The basics of the Makey Makey as an extension of the arrow keys
- The necessity of being 'connected' to the piano to play it, using a bracelet or similar item as a conductor
- How to attach crocodile clips to the board

By the end of this lesson, the children will be able to:

- Explain the fundamental workings of the bass
- Create projects in Scratch
- Enjoy playing the DIY bass guitar!



Materials Required:

- If possible, have someone bring a bass guitar and an acoustic guitar who can play them! This is a fantastic way to involve parents, older siblings, or friends.
- A printout of a guitar
- Cardboard
- Glue
- Scissors
- Markers
- Tape (easy to remove from the Makey Makey, so not too sticky)
- Aluminum foil
- Makey Makey
- Crocodile clips

Helpful links:

explanation video on the different guitars/ genres/...

- <https://www.youtube.com/watch?v=SAdk4FuE1pU>

video how to build the guitar

- <https://www.youtube.com/watch?v=0-nYfF7DHYk>

Preparation and teacher notes

If possible, ask someone to bring a bass guitar and an acoustic guitar, along with a capable player! Otherwise, you can use pictures or search online for references.

Warm up 10'

Get ready to rock out with your air guitar today!

- Stand up, strike a pose, and start jamming on your air guitar as soon as the music kicks in.
- Check out a music video or any bass compilation on YouTube for inspiration.

Describe the sensation you experience when you hear a bass guitar:

- It feels cool, groovy, ...
- Why is that? The lower tones are often used in introductions to grab your attention and get you moving!

Build-up (20 minutes):

Compare the shapes of an acoustic guitar and a bass guitar:

- Refer to Annex 1 for visuals.

A Brief History of the Bass Guitar:

- Share the history of the bass guitar with the kids, as outlined in Annex 2.

Listening Closely to Bass Guitar Examples:

- If you can connect with someone who plays the guitar, invite them to perform some tunes. If not, use a compilation video.
- Now that the kids have some background, encourage them to closely examine the instrument and listen attentively to the sound.

Explain or demonstrate that while we can play individual notes (strings), we typically use chords.

Preparation and teacher notes

It would be ideal to begin this segment after a break.

- Split the class into groups and arrange the benches. You can create as many groups as you have Makey Makey devices. Groups of three work best.

Prepare the materials for each group:

- Printout of a bass guitar
- Cardboard
- Glue
- Scissors
- Markers
- Tape
- Aluminum foil
- Makey Makey
- Crocodile clips

Building my own bass! (50')

We are going to create a simplified version of a bass that can play four notes. By touching one of the four different places on the neck (frets) we simulate the fretting and striking of a note on the guitar.

Constructing the Cardboard Guitar

1. Cut the shape of the guitar from a piece of cardboard.
2. Attach the printed drawing to your cutout (see Annex 3).
3. Place four pieces of aluminum foil along the neck of the guitar.*
4. Remember to add aluminum foil to the star, as this will serve as the input for the Makey Makey.

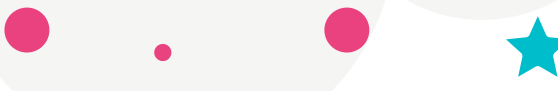
As always, there's a helpful instructional video available online. The video (starting at 6:00) may provide additional guidance. (link p. 2 helpful links)

*It would be great if the order of the strings could be specified, but the foil needs to be arranged in a loop for the Makey Makey to function properly. We understand this can be a bit confusing. (see annex 3. b)

Connecting the Makey Makey

- Refer to Annex 4 to open Scratch and build the code. If you've worked with the Makey Makey before, this is an easy task.
-

Create the code and utilize the sounds already available in Scratch. Be sure to select the BASS guitar! 😊



Preparation and teacher notes

Rehearsal Jam Session (10 minutes)

Don't forget to take some photos of your amazing class band!

Cleanup (10 minutes)

- Gently remove the tape from the Makey Makey.
- You can showcase the cardboard guitars in your classroom; they could add some fun during an music energizer activity.

Extra: Let kids ask their parents if they know a great song featuring a bass guitar. Have them share a video or audio clip on their phone or tablet, and rock out with your air guitars in the living room!

If possible, ask them to write down the song's title so we can listen to it in class the following day.



annex 1.

Feature	Acoustic Guitar	Bass Guitar
Body Size	Smaller, curvier	Larger, elongated
Neck Length	Shorter	Longer
Strings	6 (thinner)	4-6 (thicker)
Sound Production	Acoustic resonance	Deep, low-end tones

Annex 2

🎵 The Story of the Bass Guitar! 🎵

Alright, class! Today, we're going to learn about the bass guitar—the deep, groovy instrument that makes music feel full and powerful! 🎸✨

A Long Time Ago... (Before the Bass Guitar)

Before the bass guitar, people used a big instrument called the double bass (also known as the upright bass). It was HUGE—almost as tall as a person! 🎻 Musicians used it in orchestras, jazz bands, and early rock and roll, but it was hard to carry around and play on stage.

The Birth of the Bass Guitar (1950s)

🎉 In 1951, a man named Leo Fender invented the first electric bass guitar—the Fender Precision Bass! 🎸 Why was it so special?

- ✅ It was smaller than a double bass.
- ✅ It had frets, making it easier to play in tune.
- ✅ It could be plugged into an amplifier, making the sound louder!

Rock & Roll Boom! (1960s - 1970s)

The bass guitar became super popular in rock, funk, and jazz! 🎵 Famous bass players like **Paul McCartney (The Beatles)** and **James Jamerson (Motown music)** made the bass an important part of every band.

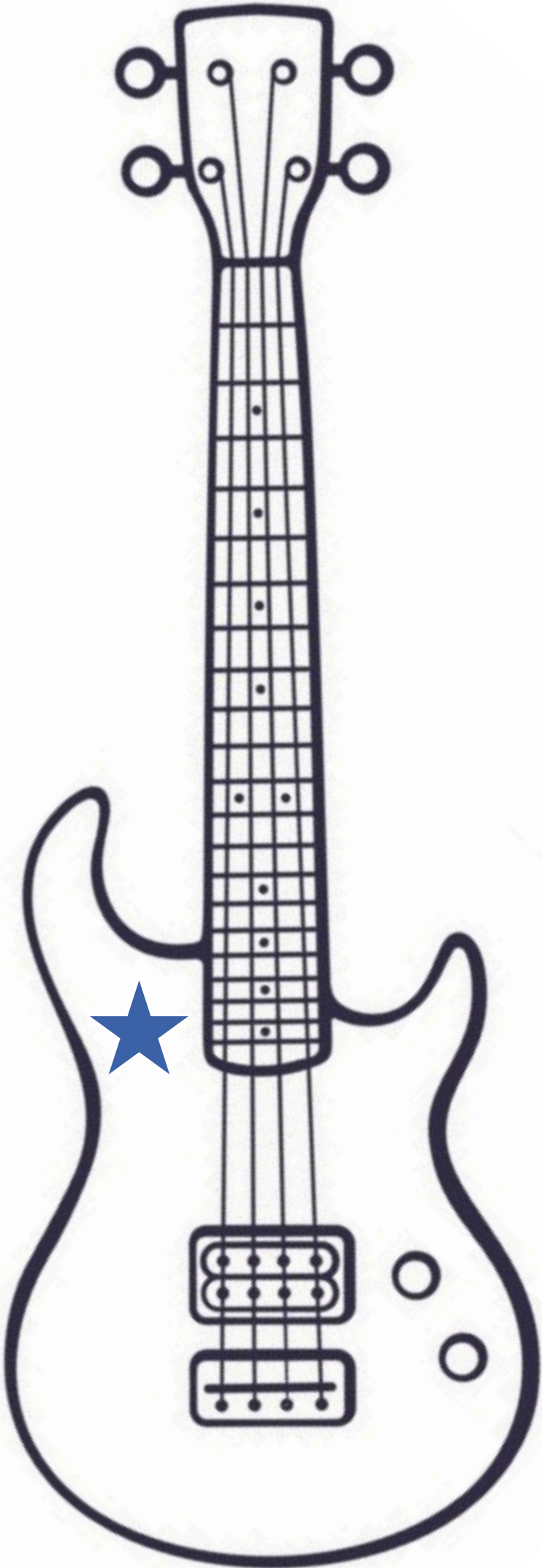
Modern Bass (1980s - Today)

Bass guitars now come in many styles—4, 5, or even 6 strings! 🎵 Amazing bass players like Flea (Red Hot Chili Peppers) and Victor Wooten show how creative you can be with the bass!

So, Why is the Bass Important?

The bass guitar is like the heartbeat of the music. It connects the drums and melody, making sure everything grooves together! Without the bass, music would feel empty and weak—so bass players are the true heroes of the band! 🧐🎸

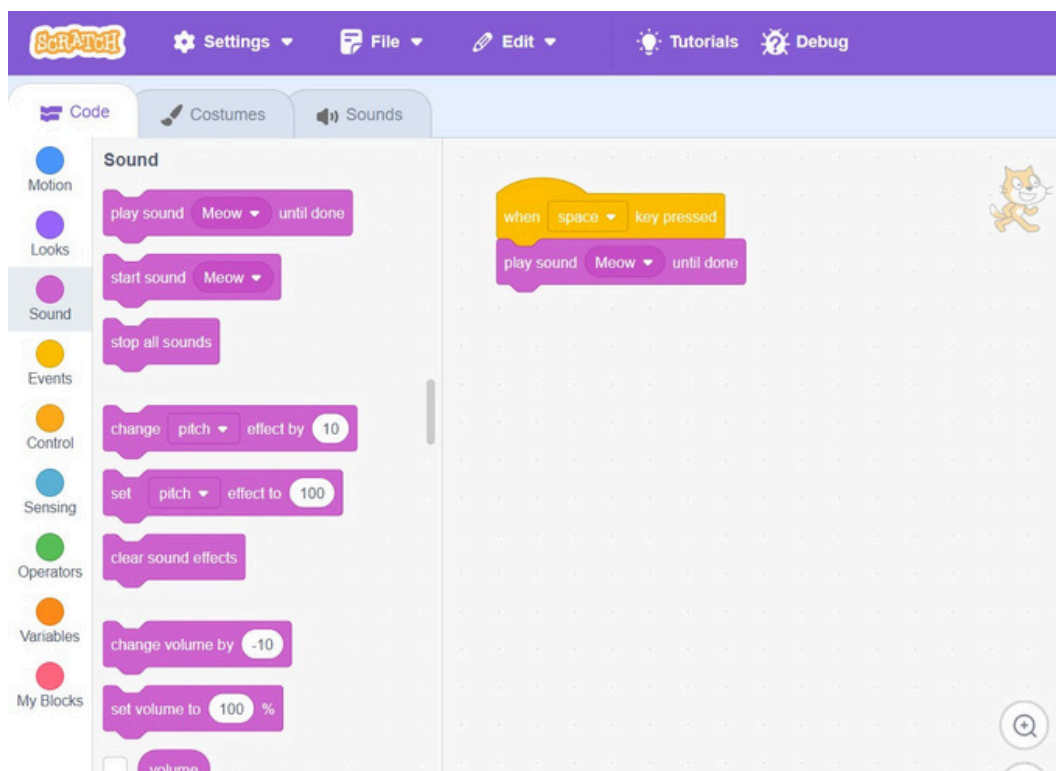
Now, who wants to try playing a bass riff? 😊🎵 Let's start to build our own bass guitar with the makey makey!



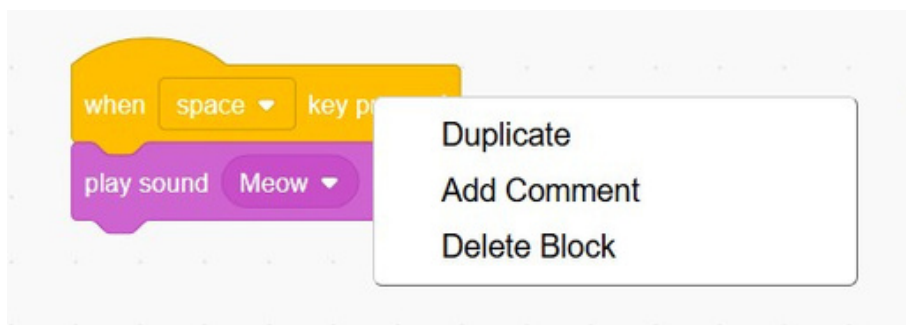


Programming Scratch

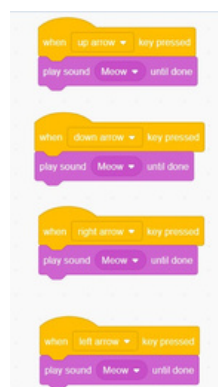
step 1: select this blocks form 'events' and 'sound'



Step 2 : duplicate this 4 times

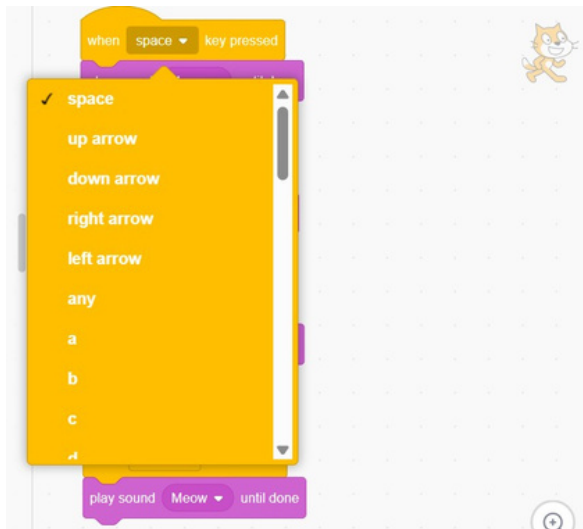


Step 3 : ready to program!



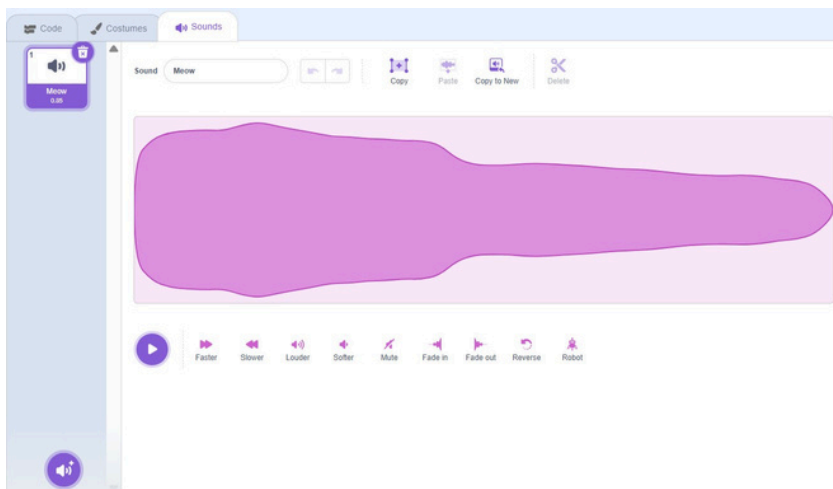
Step 4 : change to :

- when left arrow
- when right arrow
- when up arrow
- when down arrow

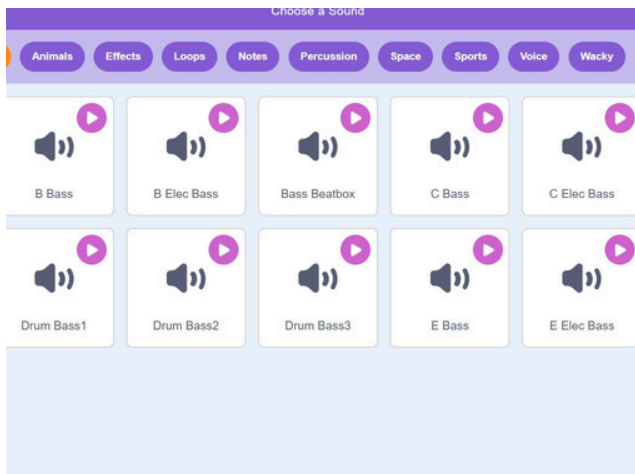


Step 5: changing the sound.

- Open sound (bottom top left)
- choose a sound (button bottom left)



Step 6: select 4 bass sound



step 7: place them in your code



Step 8: connect your makey makey!